

# GIANT CALOTROPE

*Calotropis gigantea*

## Quick Facts

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| Scientific Name | Calotropis gigantea                                                                                                                                                                                                                                           |
| Family          | Apocynaceae (Dogbane family)                                                                                                                                                                                                                                  |
| Common Names    | Giant Calotrope, Crown Flower, Madar (Hindi), Akanda (Bengali/Odia), Arka (Sanskrit), Erukku (Tamil/Malayalam), Jilledu (Telugu), Ekka (Kannada)                                                                                                              |
| Native Origin   | South Asia and Southeast Asia — India, Bangladesh, Sri Lanka, China, Malaysia, Indonesia                                                                                                                                                                      |
| Colors          | Pale purple to lilac with white. The five-petalled star-shaped corolla is white to pale purple with a purplish corona. There is also a white-flowered variety (Calotropis procera has similar white flowers).                                                 |
| Height          | 2 to 4 metres as a large shrub or small tree.                                                                                                                                                                                                                 |
| Blooming Season | Year-round throughout India. Peak flowering occurs from September to February. In West Bengal, the plant blooms prolifically from October onwards, coinciding with the Hindu festival season when the flowers are in greatest demand for religious offerings. |

## Distribution in India

One of the most widespread wild shrubs in India, found across all states from sea level to about 1,000 metres altitude. Common in wastelands, roadsides, dry riverbeds, coastal dunes, and degraded forests. Particularly abundant in West Bengal, Odisha, Bihar, Uttar Pradesh, Rajasthan, Maharashtra, Tamil Nadu, and Andhra Pradesh. Practically absent only in the extreme northeastern hill states and high-altitude Himalayan zones.

## Description and Overview

Calotropis gigantea, commonly called the Giant Calotrope or Crown Flower, is a large flowering shrub or small tree native to South and Southeast Asia. It is one of the most ecologically and culturally significant plants of the Indian subcontinent. The plant is characterised by large, thick, oval, grey-green leaves with a waxy surface that repels water. The stems and leaves contain a milky white latex that is toxic if ingested but has significant medicinal properties in traditional medicine systems.

The flowers of Calotropis gigantea are structurally complex and distinctive. Five white to pale purple petals (corolla lobes) surround a central crown-like structure (the corona) from which the plant gets the common name 'Crown Flower'. Each flower is approximately 3 to 4 cm in diameter. The fruit is a large, inflated follicle that splits at maturity to release seeds attached to long silky white fibres, which disperse on the wind.

In Hinduism, *Calotropis* holds exceptional religious significance. The flowers are considered sacred to Lord Shiva and Lord Ganesha and are offered at temples across India, particularly during Mondays (associated with Shiva) and during festivals such as Mahashivaratri. In West Bengal, the flower is offered to Goddess Kali. The leaves of the plant are also used in certain rituals.

Despite being considered a weed in agricultural contexts, the plant is of enormous ethnobotanical importance. The Ayurvedic system of medicine (the Charaka Samhita and Sushruta Samhita) describes extensive medicinal uses of practically every part of the plant — root bark, latex, leaves, flowers, and fruit. The plant is also known as 'Arka' in Sanskrit, a name that reflects its solar associations in Vedic literature.

## Blooming Season and Conditions

Year-round throughout India. Peak flowering occurs from September to February. In West Bengal, the plant blooms prolifically from October onwards, coinciding with the Hindu festival season when the flowers are in greatest demand for religious offerings.

## Cultivation

*Calotropis gigantea* is not commercially cultivated but grows spontaneously in disturbed soils across India. It thrives in dry, sandy, saline, or alkaline soils where few other plants survive, making it a pioneer species in degraded wastelands. The plant is propagated naturally by wind-dispersed seeds. It grows rapidly and can establish in challenging conditions including roadsides, coastal dunes, and railway embankments. No deliberate cultivation practices are typically applied.

## Uses and Applications

**Religious:** Flowers are offered to Lord Shiva, Lord Ganesha, and Goddess Kali in Hindu worship. The flowers, leaves, and root bark are used in various religious rituals.

**Traditional medicine (Ayurveda):** The dried root bark is used for treating skin diseases, rheumatism, bronchitis, and digestive disorders. The latex is applied externally for skin infections, boils, and toothache. Flowers are used in the treatment of asthma and catarrh.

**Fibre:** The stem bark yields a strong bast fibre used for making ropes, fishing nets, and coarse textiles in rural India. The silky seed hair (similar to silk floss) has been used for stuffing pillows and as a substitute for cotton.

**Biocontrol:** The latex contains bioactive compounds including calotropin, calactin, and cardenolides that are toxic to insects and have been studied as natural pesticides.

## Interesting Facts

- *Calotropis gigantea* is the official flower of Queen Liliuokalani of Hawaii, and the Crown Flower is the traditional flower used in Hawaiian leis.
- The plant can survive in highly saline coastal conditions where most plants fail — making it important for sand dune stabilisation.
- The silky white fibres attached to the seeds were tested during World War II as a substitute for kapok in life vests and sleeping bag insulation.

- In Ayurvedic medicine, the plant is classified as one of the most potent and toxic medicinal plants — small doses are therapeutic, large doses dangerous.
- The plant is known to survive prolonged drought by shedding leaves and entering a semi-dormant state, resuming growth when rains arrive.

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Source: Flower AI Knowledge Base — GIANT CALOTROPE entry. Scientific name: *Calotropis gigantea*. Family: Apocynaceae (Dogbane family).